

A framework for pedestrian-level wind conditions improvement in urban areas: CFD simulation and optimization

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ABSTRACT

Given the complexity of urban wind flow, acceptable wind conditions can hardly be achieved in all regions within an urban area. In some regions, high wind speeds can lead to wind discomfort and danger, while some regions suffer from poor outdoor urban ventilation. In order to evaluate and improve urban wind conditions, the combined impacts of wind flow and morphological and aerodynamic characteristics of the urban area should be taken into account. In this perspective, performing extensive and time-consuming parametric analyses is required – which is very difficult – even impossible. This paper, therefore, presents an evolutionary-based framework to optimize building heights and plan area densities to improve wind conditions in complex real urban areas. The framework consists of four steps: (i) urban area identification, (ii) optimization algorithms, (iii) CFD simulations, and (iv) evaluation. The optimization consists of Genetic Algorithm and Particle Swarm Optimization. The framework is applied to a real urban area in the city of Tehran, Iran. The focus is on minimizing low-wind-speed regions. In total, 920 possible combinations of building height variations and plan area densities are assessed. 3D steady Reynolds-averaged Navier-Stokes (RANS) simulations are performed based on a detailed validation study and grid-sensitivity analysis. The results show that the use of this framework can effectively reduce computational time and significantly improve the wind conditions in the entire urban area. Urban planners, architects, and building engineers can benefit from this framework in the early stages of design for better wind conditions in complex real urban areas.

1. Introduction

More than half of the world's population lives in urban areas [1]. The urban population will likely increase and reach 66% by 2050 [2]. As a result, urban areas are expected to get more congested, where the number of high-rise buildings will likely increase [3]. In this perspective, wind conditions in urban areas are of growing concern in contemporary society [4]. On the one hand, low wind speeds can cause near-ground air circulation limitation and entrapment of pollutants within street canyons [5,6]. This is intensified in congested urban areas [7]. On the other hand, the presence of high-rise buildings can lead to discomfort or even dangerous wind conditions at the pedestrian level [8]. Therefore, it is essential to ensure that the wind speed in urban areas remains in an acceptable range. This is, however, a challenging task.

Research on wind flow in urban areas can be studied using (i) full-scale on-site measurements [9,10], (ii) reduced-scaled wind-tunnel measurements [11,12] or (iii) numerical simulations with

computational fluid dynamics (CFD) [13,14]. Full-scale on-site measurements provide valuable data in real conditions. This approach, however, is rather expensive and time-consuming, while boundary conditions can hardly be controlled. Besides, on-site measurements cannot be performed in the early stages of design [15]. Therefore, it cannot be used for parametric or evolutionary studies [16]. Reduced-scaled wind-tunnel measurements allow a high level of control on boundary conditions. However, the use of this approach for extensive parametric studies could be expensive. Numerical simulations offer significant advantages over the other methods. They are less expensive than experimental methods, and they provide information at every point under fully controlled boundary conditions. However, the accuracy and reliability of simulation results need to be assessed accurately and precisely [17].

Detailed evaluation and improvement of wind conditions in urban areas is very complicated because urban wind flow depends on different parameters, including urban area density [18,19], building geometry [14,20,21], street canyon geometry [22–24], and wind speed

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Nomenclature

Abbreviations

Aerodynamic Index of Urban Area AIUA
 Evolutionary Algorithm EA
 Genetic Algorithm GA
 Hybrid of GA and PSO HGP
 Particle Swarm Optimization PSO

Symbols

Height of one floor (= 2.8 m) H
 Longitudinal turbulence intensity (%) I_u
 Von Karman constant (0.42) κ
 Land length (m) L
 Building length (m) l
 Streamwise wind velocity (m/s) u
 Atmospheric Boundary Layer friction velocity (m/s) U_{ABL}^*
 Building height (m) z

characteristics [25,26], etc. A complete understanding of this complexity, therefore, highly depends on the detailed information about the individual and combined impacts of these parameters, which are not easily varied independently. This is a multi-criteria problem, and hence, it would be very time-consuming, and even impossible, to perform typical parametrical studies [27]. Therefore, evolutionary optimization techniques, which are efficient alternatives to classical methods [28], should be pursued.

The use of evolutionary optimization goes back to the 1950s [28]. These methods can be used effectively in the most complicated problems that seek optimization of a particular multi-criteria function [29]. In nature, evolution is the ability of self-modification, where it changes in the characteristics of a species over generations in its environment [30]. Evolutionary algorithms (EAs), which mimic the way natural systems survive [27], have been entered various architectural fields and studied by several researchers (e.g. [31–37]). This approach has been successfully used on several occasions in the past for different applications. For example, coupled EAs and CFD simulations were been performed for indoor climate conditioning [38] and coupled EAs and Response Surface Model were been employed in order to lift-up design of an isolated building [39]. However, despite the abovementioned advantages, EAs and CFD simulations have not yet been implemented in a coupled fashion to improve the wind conditions in complex urban areas.

This paper, therefore, presents an evolutionary-based framework to optimize the morphological characteristics of a complex real urban area in order to improve the wind conditions. The framework consists of four steps: (i) urban area identification, (ii) optimization algorithms, (iii) CFD simulations, and (iv) evaluation. The presented framework aims to investigate the combined impact of building height and plan area density variations on wind conditions.

The optimization includes a hybrid of the Genetic Algorithm (GA) [40] and Particle Swarm Optimization (PSO) [41] to reach optimized patterns for building heights and plan area densities in complex real urban areas. The new algorithm, which is a hybrid of GA and PSO, is developed in Python based on the algorithms provided in Refs. [42–44]. Note that in the remainder of this paper, the term “HGP” stands for the “Hybrid of GA and PSO” algorithm.

It is well known that the use of PSO, compared with GA, can lead to more optimal solutions in fewer iterations. However, the Genetic Algorithm ultimately yields a better result in the same number of evolutionary periods [27,45,46]. In GA, the information contained by an individual is lost if it is not selected, but PSO has memory to record data related to individuals. However, PSO is very likely to waste memory on poor individuals in the absence of GA operators [47]. The newly

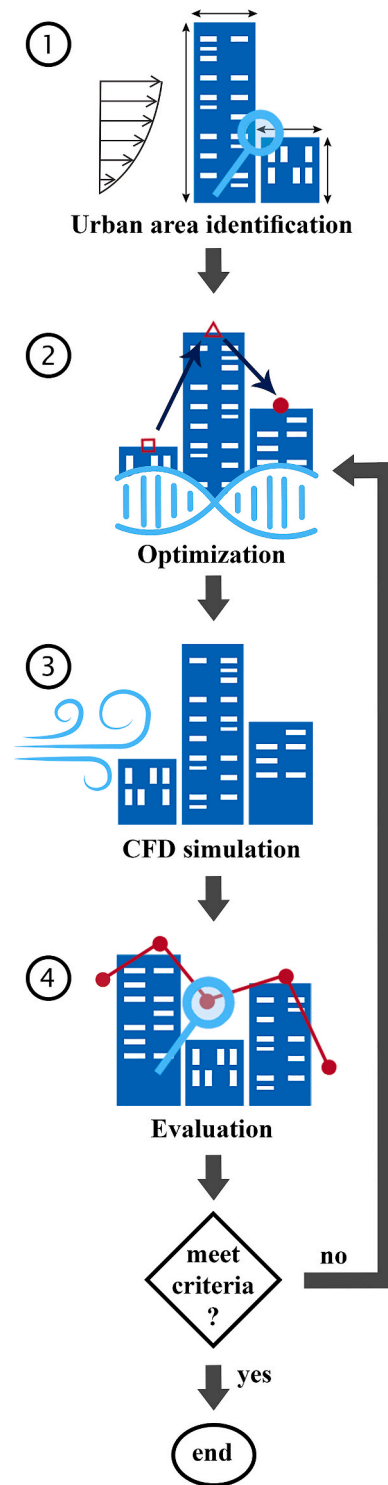


Fig. 1. Schematic diagram for steps of proposed framework.

developed algorithm (HGP), therefore, can get leverage over both the abovementioned algorithms in multi-criteria problems. For example, the HGP can be efficiently used in wind comfort assessment for complex real urban areas, where reducing the computational time and cost of simulations is of growing concern.

The framework is also applied to a real urban area located in the city of Tehran, Iran. The focus is on minimizing low-wind-speed regions. In total, 920 possible combinations of building height variations and plan area densities are assessed. 3D steady Reynolds-averaged Navier-Stokes

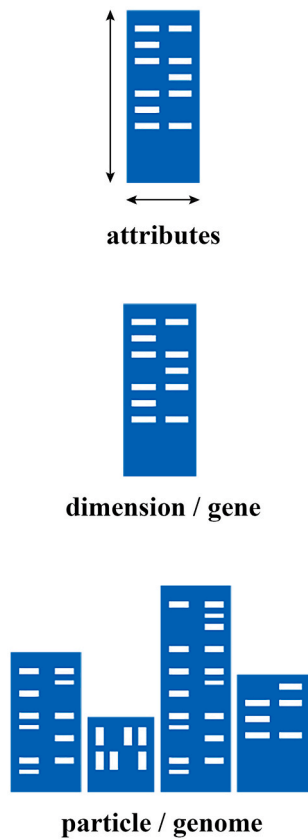


Fig. 2. Difference between attributes, dimension (gene), and particle (genome).

(RANS) simulations are performed based on a detailed validation study and grid-sensitivity analysis.

Such a framework can be a valuable addition to the literature. Urban planners, architects, and building engineers can benefit from this framework in the early stages of design to improve wind conditions in complex real urban areas.

In Section 2, the proposed framework is described. Section 3 presents the application of the framework in a real complex urban area. A critical assessment of the framework is given in Section 4. A discussion on the limitations of the study and the main conclusions are presented in Section 5 and Section 6, respectively.

2. Description of framework

The procedure of the proposed framework, which is a combination of evolutionary optimization and CFD simulation, consists of four main steps, as illustrated in Fig. 1.

2.1. Step 1: urban area identification

In the first step, some main specifications of the area of interest need to be initially identified. It includes the number of buildings, building heights, plan area densities, and wind characteristics. The morphological and meteorological information for real urban areas can be collected through field observations, master plans, meteorological stations, etc. This information can be collected for the area of interest and surrounding buildings. It should be noted that surrounding buildings are normally included in CFD simulations of wind flow in urban areas to better provide accurate results [48]. However, the number of selected buildings and the extension of the area of interest highly depends on the availability of computational resources. Finally, these specifications are used as input parameters for the following steps.

Table 1
Schematic of the optimization method.

Operator	Scheme	Application
PSO ^a		
Crossover ^b		
Mutation ^c		

^a circle indicates previous position, square indicates global best position, triangle indicates personal best position and filled circle indicates current position.

^b filled circle indicates parent X, square indicates parent Y, and combined shapes indicate offspring x and y.

^c filled circle indicates parent X and rectangle indicates offspring x.

2.2. Step 2: optimization

In the second step, EAs are employed to improve wind conditions for a certain level(s), i.e., at the pedestrian level. The hyper-parameters of the problem, which are the population size (the number of generated urban areas in each period), the number of generations, the crossover rate, and the mutation rate, are decisions made by the user according to the available facilities. In this framework, the termination condition is defined as reaching an absolute number of generations, which highly depends on the time and cost of simulations. In the case of the availability of computational resources, the number of generations can be increased. The data from Step 1, together with these hyper-parameters, are used in this step to perform an optimization.

From an evolutionary standpoint, generated urban areas are just “particles” (in PSO) or “genomes” (in GA) of the problem, and the number of involving buildings is the number of “dimensions” (in PSO) or “genes” (in GA). Building properties (i.e., building height and plan area density) are also known as the attributes of the dimensions or genes. Fig. 2 shows the difference between attributes, dimension (gene), and particle (genome). In the first evolutionary period, a population of urban areas is randomly initialized. The ranges for each attribute are based on field observations. Note that this is an iterative process in which simulation continues until the termination condition is met.

Table 1 illustrates how different parts of the HGP algorithm work and interact. In the PSO, each possible solution, which is called a particle, consists of several buildings (the number of dimensions). There are some decisive factors, including deciding on the number of dimensions per particle and specifying height and plan area density ranges for the buildings (attributes). Each particle deals with a phenomenon called velocity updating, which is the vector of motion towards the optimal solution at the current evolutionary period. The position and the velocity of the particles are obtained using Eqs. (1) and (2), respectively [49], where C_1 shows the impact of the particle experience, C_2 shows the impact of other particle experiences, and $Rnd(0.1)$ is a uniform random number between 0 and 1. Accordingly, the position of the particle i with dimension d is updated ($P_{i;d}(it + 1)$) in each iteration (it) using its corresponding velocity ($V_{i;d}(it + 1)$), which depends on the global optimum ($gb_d(it)$), the local optimum ($pb_{i;d}(it)$) of the particle, and the previous position of the particle ($P_{i;d}(it)$).

$$P_{i;d}(it + 1) = P_{i;d}(it) + V_{i;d}(it + 1) \quad (1)$$

$$\begin{aligned} V_{i;d}(it + 1) = & V_{i;d}(it) \\ & + C_1 * Rnd(0,1) * [pb_{i;d}(it) - P_{i;d}(it)] \\ & + C_2 * Rnd(0,1) * [gb_d(it) - P_{i;d}(it)] \end{aligned} \quad (2)$$

In the GA, each generated urban area is a genome that represents a possible solution. The genomes contain several buildings or genes. Each gene has three attributes, including height, plan area density, and the criteria of the algorithm for global optimization. Like PSO, there are some decisive factors in the GA part. They are deciding on the number of buildings (genes) in each generated urban area (genome) and assigning a range for height and plan area density (attributes). Blend Crossover Algorithm [42] is employed for applying crossover, which consists of the following steps:

1. Select two parents $X^{(t)}$ and $Y^{(t)}$ from the parent pool
2. Create two offspring $x^{(t+1)}$, and $y^{(t+1)}$ as follows:
3. For $i = 1$ to n do
4. $\Delta = 0.2$
5. Choose a uniform random real number $\alpha \in \langle -\Delta, 1 + \Delta \rangle$
6. $x_i^{(t+1)} = (\alpha_i \times X_i^{(t)}) + ((1 - \alpha_i) \times Y_i^{(t)})$
7. $y_i^{(t+1)} = (\alpha_i \times Y_i^{(t)}) + ((1 - \alpha_i) \times X_i^{(t)})$
8. End do

For the mutation part, Point Mutation algorithm [44] is used, and the

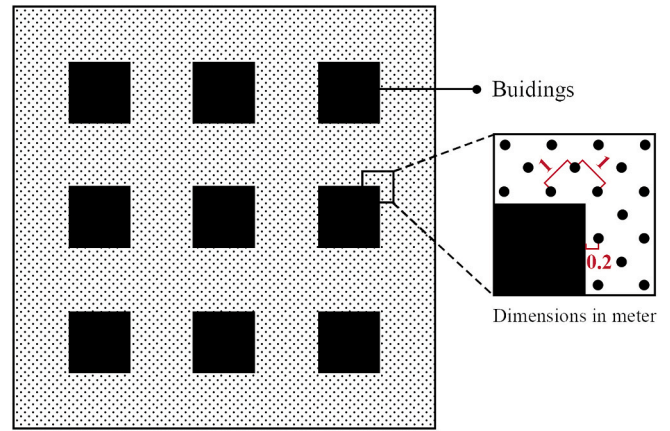


Fig. 3. Schematic of buildings and points for which wind speed is recorded.

following steps are given (Δ refers to the parents' attribute range):

1. Select one parent $X^{(t)}$ from the parent pool
2. Create one parent $x^{(t+1)}$ as follows:
3. Choose an integer random real number $u \in 0, n - 1$
4. $\sigma = \Delta/10$
5. Choose a Gaussian random real number $g \in 0, 1$
6. $x^{(t+1)} = X_u^{(t)} + (\sigma \times g)$

2.3. Step 3: CFD simulation

In the third step, CFD simulations are performed for the urban areas generated using the optimization algorithm in Step 2. This step should be made with special attention to the choice of numerical models, performing a validation study and a grid-sensitivity analysis, and the use of a relevant performance indicator.

• Numerical model

In this study, 3D steady Reynolds-averaged Navier-Stokes (RANS) CFD simulations are performed for generated urban areas.

• Validation study and grid-sensitivity analysis

For CFD simulations, the accuracy and reliability of results are of significant concern. Depending on the level of complexity of the area of interest, and availability of computational data, a validation study for the same area or a sub-configuration validation will be performed. In addition, in order to ensure that the CFD results are independent of the computational grid, a grid-sensitivity analysis need to be performed. Since the optimization process is automatically done, the validation study and grid-sensitivity analysis are not included within Step 3, as they are performed only once before the optimization process starts.

• Performance indicator

A performance indicator should be introduced to continue the iterative process. This indicator aims to evaluate the capacity of a real urban area to dilute and remove pollutants. The Aerodynamic Index of Urban Area, called AIUA, is used as the output of the fitness function of the EA. The AIUA is to be minimized over evolutionary periods and is calculated as follows:

For each generated urban area, a horizontal plane at the pedestrian level (1.75 m) is considered. The plane contains uniformly distributed points, as shown in Fig. 3. The distance between all points is considered to be 1 m. Note that in some positions, such as near building walls, a smaller distance needs to be considered.

The wind speed at each point is then measured through CFD simulation and evaluated using an acceptable range for mean wind speed. For the upper bound of the range, the mean wind speed is considered to be less than 5 m/s, which is in line with the Dutch standard for wind comfort and wind safety (NEN 8100) [50] but also suggested in Ref. [51]. For the lower bound, the mean wind speed is considered to be higher than 1 m/s, which is in line with the suggestions by Ref. [51]. Therefore, the outdoor wind speed that provides the most comfortable environment at the pedestrian level is considered to be between 1 m/s and 5 m/s, which underpins the decision making in this framework. Lower wind speeds may cause problems with outdoor urban ventilation, and higher wind speeds surpass wind comfort criteria. Accordingly, each point within the area contains a value to better evaluate the wind conditions. In this perspective, if the wind speed at a point is within the acceptable range, the corresponding value is zero. For wind speeds outside the acceptable range, the absolute difference between them and the upper/lower bound is considered as the value.

For instance, if the wind speed at a point is 0.5 m/s, the absolute difference between 0.5 m/s and 1 m/s (lower bound) is 0.5 m/s. Thus the value for the point is 0.5. Likewise, the value for a wind speed of 5.5 m/s is equal to 0.5, as the absolute difference between 5.5 m/s and 5 m/s (upper bound) is also 0.5 m/s. Finally, the values of all points are added up, and the AIUA is computed using Eq. (3), where v_i refers to the wind speed at each point, v_c is the upper/lower bound of the range of the acceptable wind speed and n is the number of points at the horizontal plane. Note that the AIUA is the accumulative sum of the difference between the wind speed and the closest bound of the acceptable wind speed range for all points on the horizontal plane.

$$\sum_{i=1}^n |v_i - v_c| \quad (3)$$

The ideal AIUA that corresponds to perfect wind conditions in the area of interest is zero. Meaning that wind speeds at all points are within the acceptable range. Since an AIUA = 0 is challenging to achieve, efforts should be made to keep this value as small as possible. It also should be noted that lower AIUA values lead to more favorable wind conditions in urban areas in view of both outdoor urban ventilation and wind comfort improvement.

In each evolutionary period, after assigning the attributes (height and plan area density) to the buildings (Step 2) and applying CFD simulation for each urban area (Step 3), the AIUA is calculated. The population is then sorted based on the lowest value of the AIUA up to the current evolutionary period, and it is sent to the next evolutionary period until the termination condition is met. Finally, the optimal case is sent to Step 4 for evaluation.

2.4. Step 4: evaluation

In the fourth step, the optimal urban area is evaluated. Some situations would happen that a favorable result is not obtained when the termination condition is met. Extending the number of generations, and tuning the hyper-parameters to re-run the simulation are among actions that can be taken in this regard.

3. Framework application

According to Iran National Building Regulations [52], buildings within an urban area are of a particular maximum height and a specific plan area density. Such restrictions result in having a steady skyline in most urban areas in Iran. However, previous studies have shown the important impact of height variation on wind conditions in urban areas [6]. In order to illustrate the potential of the proposed framework to improve wind conditions in complex real urban areas, a case study is presented for a real urban area in the city of Tehran, Iran.

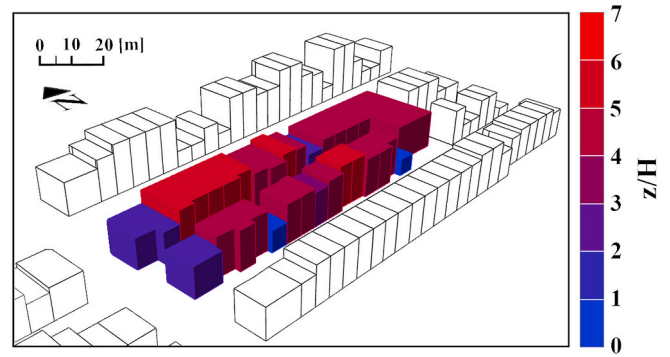


Fig. 4. Perspective view of case study with indication of building heights.

3.1. Step 1: urban area identification

The area of interest is a complex real urban area consisting of 32 residential low-rise buildings located in the city of Tehran, Iran. Based on the classification provided in Ref. [53], the area under study is classified as Local Climate Zone 3 (compact low-rise buildings). It also suffers from air pollution issues. Fig. 4 shows a perspective view of the area of interest and its surrounding buildings (in white color). For each building, the average width and length are about 7 m and 12 m, respectively, and the number of floors ranges from 1 to 7. All streets are double direction for vehicles with width ranging from 6 m to 20 m. The aspect ratio of the street canyons ranges from 0.5 to 1.

The wind statistics of the year 2017 from a meteorological station, 10.6 km far from the case study [54], is used (Fig. 5). In this case, the mean wind speed at $H = 10$ m is 2.96 m/s and the prevailing wind direction is $\theta = 299^\circ$ from North. Considering the complexity of urban geometry and the high computational cost, only one wind direction is used in this study. Fig. 5 shows a plan view of wide surroundings with a radius of 11 km around the case study. The central black rectangle indicates the computational domain for the case study. According to the updated Davenport-Wieringa roughness classification [55], the aerodynamic roughness length (z_0) is estimated to be 1 m for the prevailing wind direction.

3.2. Step 2: optimization

The size of the initial population is 20 particles, and the number of evolutionary periods is considered to be 25 iterations. 100% of the population (20 generated urban areas) undergo the PSO operators. Afterward, 60% of the population (12 generated urban areas) participate in the crossover. Finally, 20% of the population (4 generated urban areas) enter the mutation process to create new parents. Considering the computational cost of the optimization process, there is a limitation on the size of the population and the number of iterations. Therefore, the mutation rate is increased by up to 20% to yield a far superior result [56]. Thus, $25 \times 36 = 900$ cases are generated during the evolutionary periods. Together with the initial population, 920 urban areas are generated in total.

Each urban area holds 32 buildings. It means that each particle or genome in the population has 32 dimensions or genes. Moreover, each building contains two attributes of building height and plan area density. In the initial phase of the simulation and before the evolutionary period starts, these two attributes are randomly assigned. The ranges for plan area densities and floor numbers are 50%–65% and 1–7, respectively. Fig. 6 shows the initial population for the case study. After each iteration, the generated population through PSO, crossover, and mutation operators (Table 1) are combined, sorted, and sent to the next evolutionary period. The termination condition for the optimization process is considered to reach 25 generations. Thus, after 25 periods and generating 920 urban areas, the simulation ends.

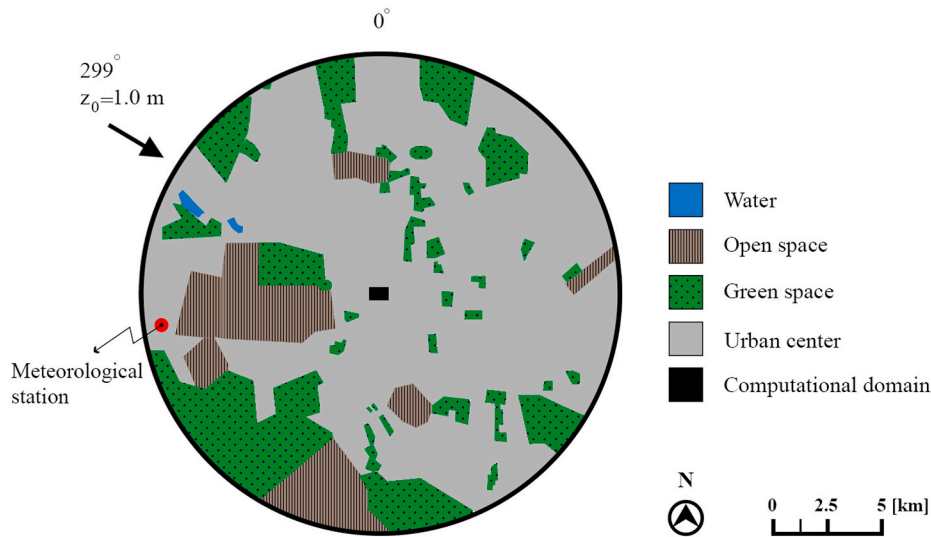


Fig. 5. Surroundings of case study in an 11 km radius with the aerodynamic roughness length z_0 for prevailing wind.

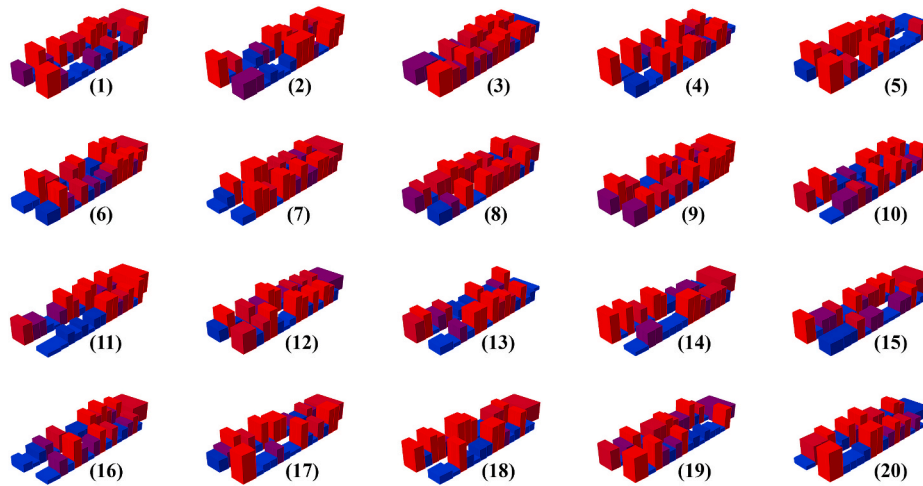


Fig. 6. Randomly generated initial population of 20 particles with indication of building heights. See Fig. 4 for color bar. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.3. Step 3: CFD simulation

In the third step, CFD simulations are performed for the generated urban areas to record the wind speed at the pedestrian level and consequently calculate the AIUA. It should be noted that a detailed validation study for RANS simulations is initially performed.

3.3.1. Computational domain, geometry, and grid

A computational domain of the urban area and surrounding buildings is generated based on the best practice guidelines [57–61]. The height of the domain from the top of the buildings and the downstream length are $6H_{\max}$ and $10H_{\max}$, respectively, where $H_{\max}$ is the maximum building height in the case study ($H_{\max} = 14\text{m}$) (Fig. 7a). In addition, the distance between the area of interest and inlet and lateral planes are $6H_{\max}$, which is in line with the abovementioned guidelines (Fig. 7b). It should be noted that the length of the domain relative to the surrounding buildings is about $2H_{\max}$. The choice of this rather small length is to limit the deterioration of the inlet profiles as has also been successfully conducted in previous CFD studies [62]. In order to ensure that the upstream and lateral lengths of the domain have an insignificant impact on the simulation results, a sensitivity analysis is performed for two

upstream and lateral length values: $2H_{\max}$ and $5H_{\max}$. The results show that the overall difference in the wind speeds along two lines within the domain is less than 5%.

The geometry of the surrounding buildings is modeled in a radius of 20 m (almost two times the average height of the buildings in the urban area [60]). The buildings are also simplified as rectangular prism without any openings to reduce the computational cost. Note that earlier studies have shown that the geometrical details of a building could significantly impact the near-wall flow patterns [63].

As shown in Fig. 7, the inlet wind profiles are perpendicular to the inlet plane. Since an automatic procedure has been used for the grid generation, it was found that such a computational grid, while the inlet profiles are perpendicular to the inlet plane, could contribute to reducing the overall computational time. That is extraordinarily important given the fact that several iterations should be done until optimum results are obtained.

The computational grid is generated with the aid of pre-processor components of Butterfly, which are connected to the blockMesh and snappyHexMesh tools of OpenFOAM. According to the best practice guidelines for grid generation [57,58], a systematic and detailed grid-sensitivity analysis is performed. Two additional grids are

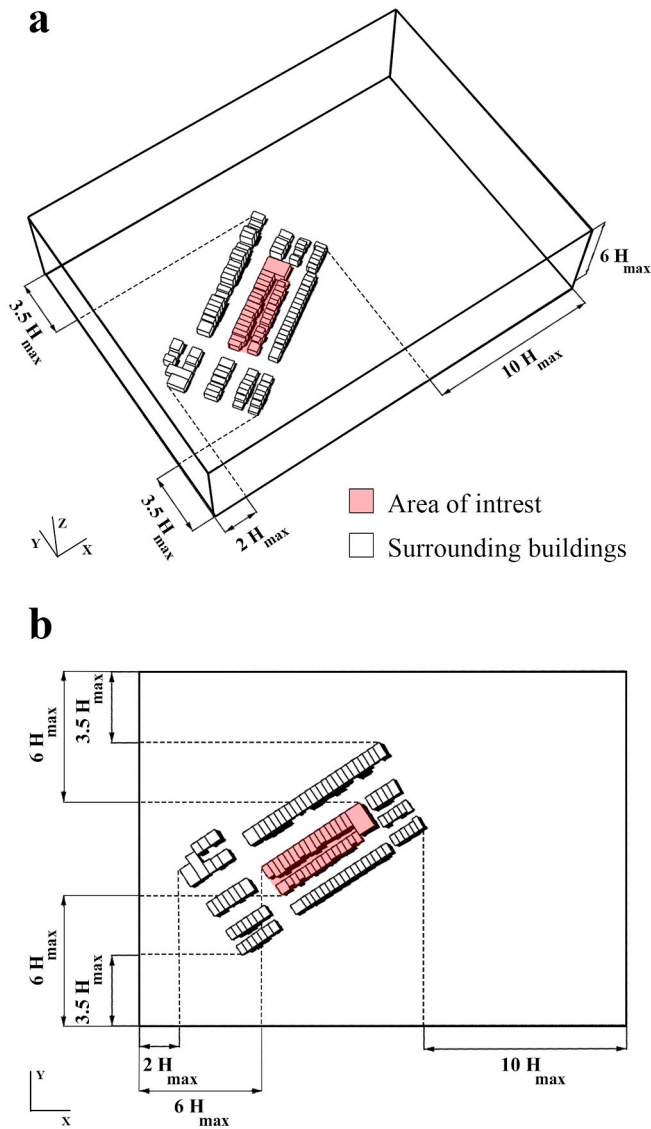


Fig. 7. Computational domain: (a) Perspective view, (b) Top view of actual upstream and lateral length for area of interest.

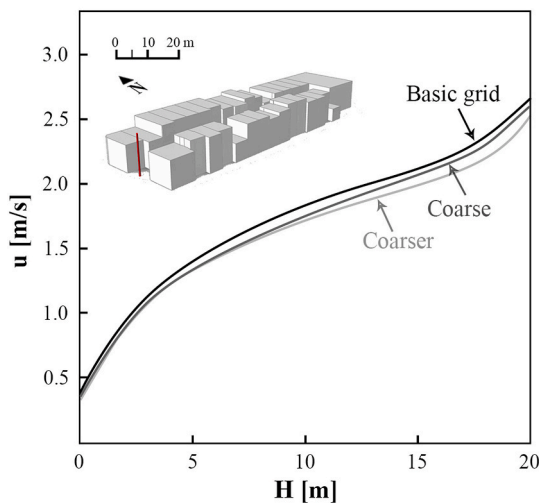


Fig. 8. Grid-sensitivity analysis: streamwise wind speed along a line for the three grids.

Table 2
Boundary conditions.

Boundary type	Conditions
Inlet	Wind speed of 2.99 m/s is considered at the height of 10 m
Ground and building walls	No-slip boundary conditions
The lateral and top boundary	Symmetry boundary conditions
Outlet	Zero gradient pressure

generated: a coarse and a coarser grid. Coarsening is performed with an overall linear factor of $\sqrt{3}$. The total number of cells is 213,020 for the coarser, 371,762 for the coarse, and 610,515 for the basic grid. Fig. 8 shows the streamwise wind speed along a line in the middle of the urban area for the three grids. The comparison of the results of the basic and coarse grids and the coarse and coarser grids show an average absolute deviation of 2% and 4%, respectively. Therefore, the basic grid is retained for all the simulations. The maximum height of the first cell adjacent to the building is a maximum size of 0.46 m [59], which is in line with the recommendations by the best practice guidelines for having at least two or three layers of the control volume below the pedestrian level [61]. This results in a blockage ratio of 3%, which is also in line with the guideline mentioned above.

3.3.2. Boundary conditions and solver settings

The boundary conditions for computational domain are described in Table 2. Neutral atmospheric boundary layer inflow of mean wind speed (Eq. (4)), turbulent kinetic energy (Eq. (5)), and turbulent dissipation rate (Eq. (6)) are imposed at the inlet of the domain. A detailed description of inlet boundary conditions is available in Ref. [63]. The aerodynamic roughness length for the ground and building walls are modeled as smooth walls with $z_0 = 0$. For the outlet of the domain, zero gradient pressure is imposed.

$$U(z) = \frac{u_{ABL}^*}{\kappa} \ln\left(\frac{z + z_0}{z_0}\right) \quad (4)$$

$$k(z) = \alpha(I_u(z)u(z))^2 \quad (5)$$

$$\varepsilon(z) = \frac{u_{ABL}^{*3}}{\kappa(z + z_0)} \quad (6)$$

The simulations are performed using Butterfly, which is a translation of OpenFOAM open-source software into Grasshopper [64]. The 3D steady Reynolds-averaged Navier-Stokes equations are solved in combination with the standard $k-\varepsilon$ turbulence model, which is selected based on the detailed validation study provided in Section 3.3.3. Convergence is obtained when all the scaled residuals reach a minimum of 10^{-6} .

3.3.3. CFD validation

In this study, the validation is based on the wind-tunnel measurements of mean wind speed for a generic urban area by Nonomura et al. [65]. In the experiment, the mean wind speed was measured for three reduced-scale generic urban areas (1:100) and three wind directions. In the present validation study, the focus is on a group of nine identical cubic buildings with a height of 0.2 m (Fig. 9a) under $\theta = 22.5^\circ$, which is more relevant for the present study.

The measured and simulated mean wind speeds are compared for all 120 measurement points in the experiment. All these points are located at the pedestrian level. Fig. 9 shows the results along two horizontal lines $X = 0.2$ m (Fig. 9b) and $Y = -0.2$ m (Fig. 9c). In Fig. 9, $U_{ref} = 2.43$ m/s (wind speed at $H = 0.02$ m) is used for normalization.

A fairly good agreement can be seen between the CFD results and the measured data. The average and the maximum deviations are about 3% and 8%, respectively. The main reason for these deviations is not clear, but it could be because the steady RANS is not capable of reproducing

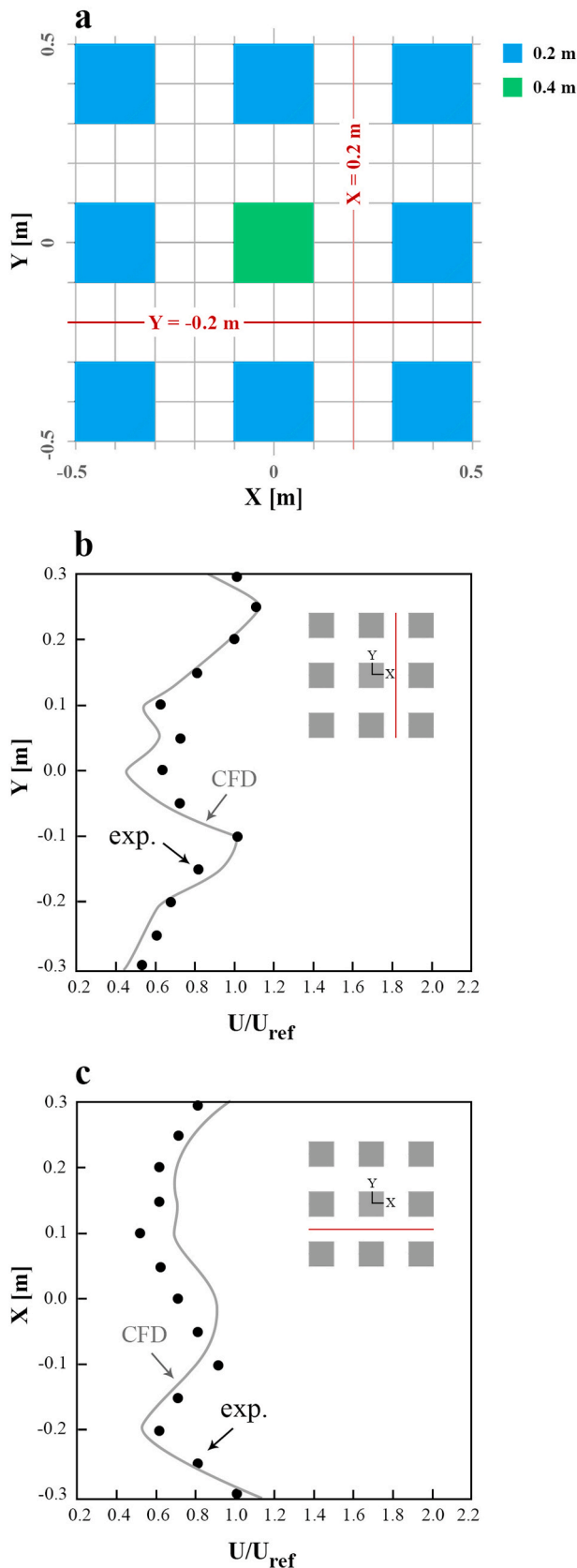


Fig. 9. CFD validation: (a) Top view of reduced-scale generic urban area. Comparison of CFD results and experimental results of mean wind speed along lines (b) $X = 0.2$ m and (c) $Y = -0.2$ m.

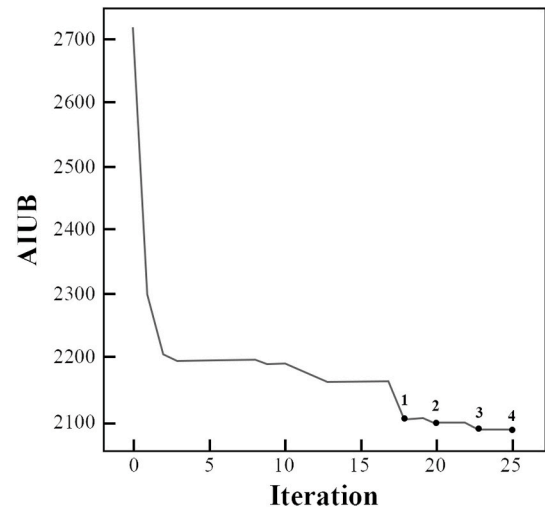


Fig. 10. AIUA of optimal generated urban area in each evolutionary period. Numbers indicate Case 1–4.

the separation and recirculation downstream of the windward façade [11,15]. On the other hand, the standard $k-\epsilon$ model is not able to predict the recirculation on the roof and upstream dispersion [16].

3.4. Step 4: evaluation

The termination condition for the optimization process is to reach 25 generations in order to limit the computational time and cost. Fig. 10 shows the best solution reported in each evolutionary period. According to this figure, the best AIUA obtained after 25 periods (Fig. 10, point 4) is improved by about 19% compared to the current situation of the case study (Fig. 11a), resulting in Case 4 as the final optimum solution (Fig. 11b). Note that for comparison purposes and to provide insight into the performance of the framework, the results of Case 1–3 (Fig. 10, points 1–3) are also provided in the following Section.

The AIUA for the case study at the pedestrian level is 2570, and the total area of its buildings is 12190 m^2 . The number of floors and plan area density for most buildings are 4 and 60%, respectively. As shown in Fig. 11a, in areas like between two rows of buildings, the wind speed is less than 0.5 m/s for the most part. These low-wind-speed regions are very likely to suffer from air pollution stuck. Fig. 11b shows the optimal solution obtained through the application of the framework. The AIUA for the optimal case is 2087, which is considerably improved compared to the current situation. According to Fig. 11, building heights are highly varied, and hence, low-wind-speed regions are significantly reduced.

4. Critical assessment of framework

In this study, EAs are employed to optimize a single-objective problem (i.e., minimizing the AIUA). However, a critical analysis is presented here to evaluate the proposed framework. Therefore, in order to provide insight into the performance of the framework, the following investigations are given:

- The top four generated urban areas are investigated in Fig. 12 and Table 3.
- The effect of GA and PSO operators are analysed separately in Fig. 13 by comparing the building height variation in three cases.
- The AIUA is calculated for a vertical plane between two rows of buildings within the case study, and the results are presented in Table 4.
- The amounts of the total area of living are compared in Table 5.
- The heights distributions for all cases are shown in Fig. 14.

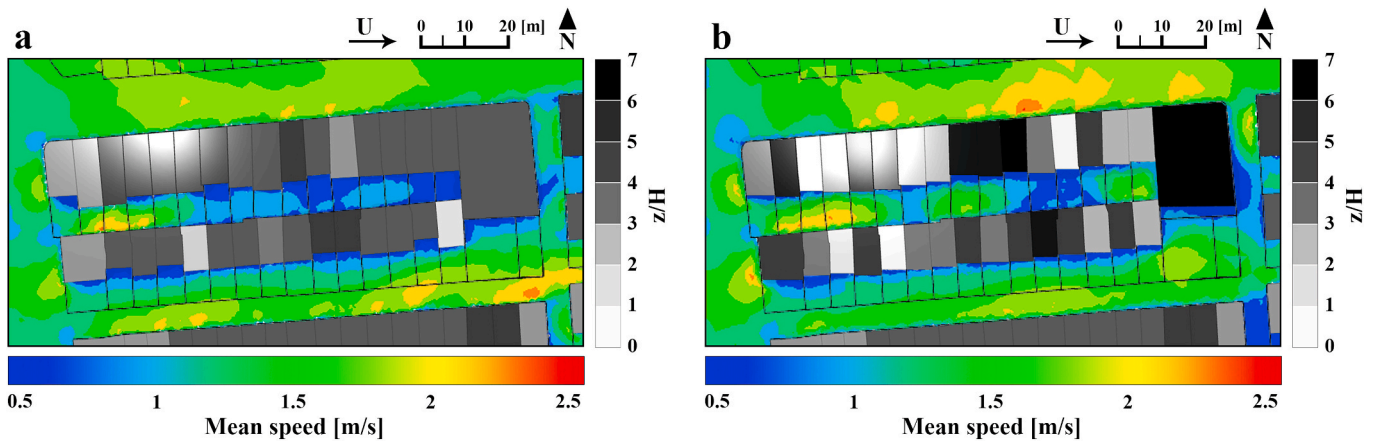


Fig. 11. Mean wind speed distribution at pedestrian level (1.75 m) for (a) Original case study, (b) Optimal solution. Gray color bar indicates building heights. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

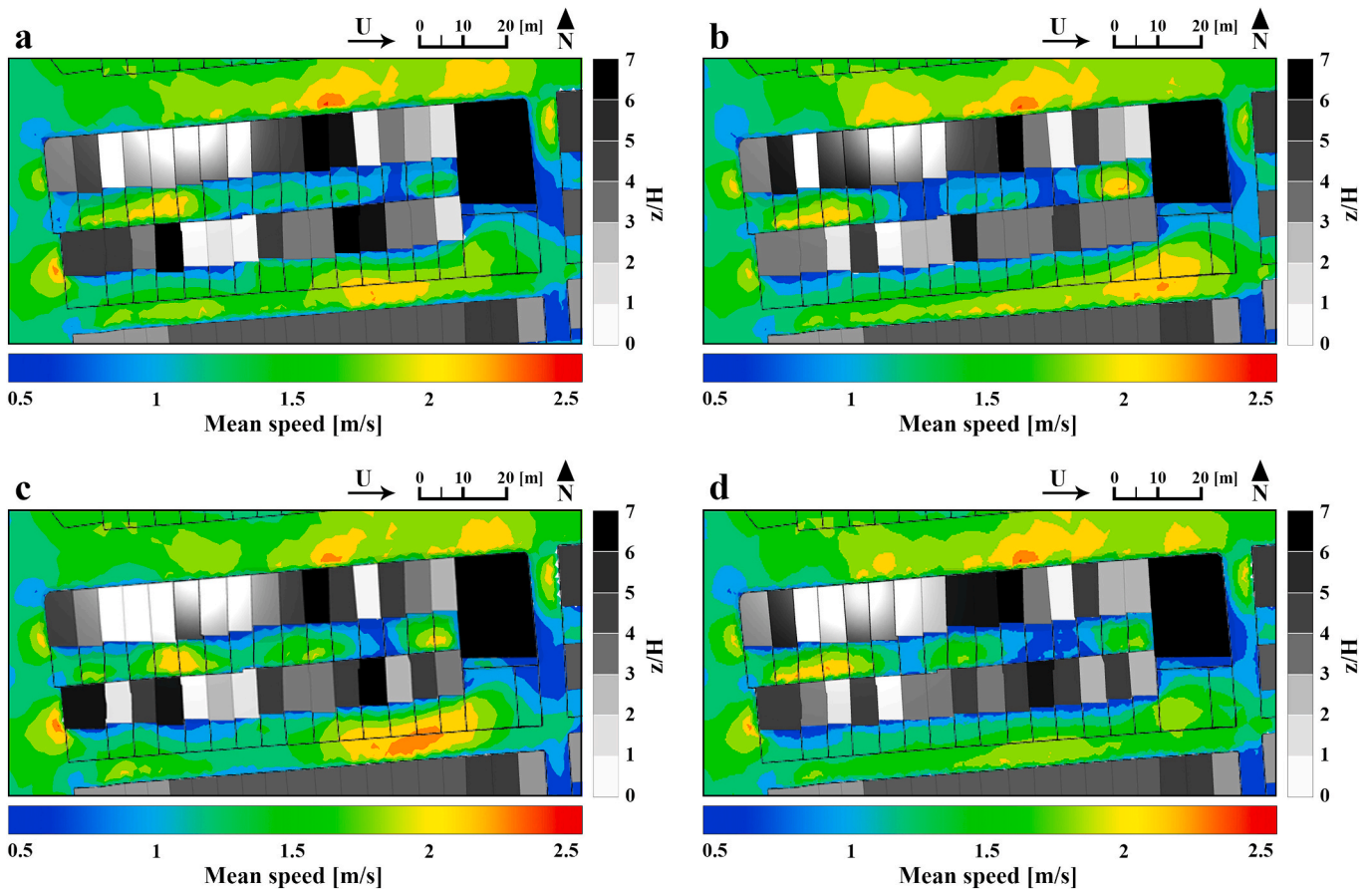


Fig. 12. Mean wind speed distribution at pedestrian level (1.75 m) for (a) Case 1, (b) Case 2, (c) Case 3, and (d) Case 4. Gray color bar indicates building heights. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

- The distributions of plan area densities for all cases are presented in Fig. 15.

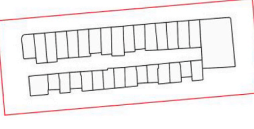
Case 1 (Fig. 10, point 1) is the best solution to the 17th generation with the AIUA of 2163 and a total area of 12045 m^2 . The maximum number of floors and the maximum plan area density are 4 and 58%, respectively (Fig. 12a). Case 2 (Fig. 10, point 2) is the best solution to the 20th generation, with the AIUA of 2104 and a total area of 12221 m^2 . The figures for the maximum number of floors and the plan area density are the same as Case 1 (Fig. 12b). Case 3 (Fig. 10, point 3) is the best

solution to the 23rd generation with the AIUA of 21,013 and a total area of 12002 m^2 . The maximum number of floors is 5, and the maximum plan area densities are 56% and 58% (Fig. 12c). Case 4 (Fig. 10, point 4) is the optimal solution, with the AIUA of 2087 and a total area of 12302 m^2 . The maximum number of floors and plan area density for Case 4 is 5 and 57%, respectively (Fig. 12d).

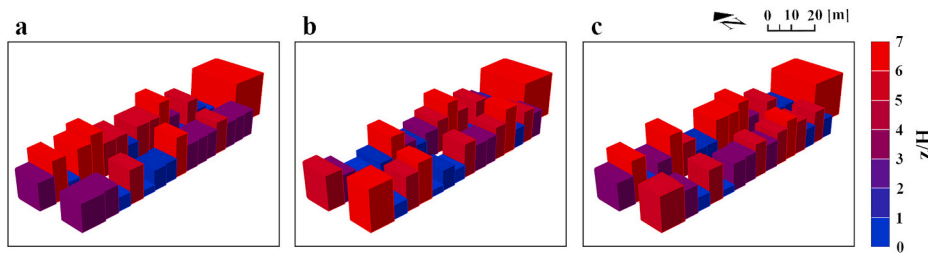
Given the need to evaluate the capacity of the generated urban areas to improve wind conditions, the AIUA (Aerodynamic Index of Urban Area) is first introduced in this study. Other factors are also considered in Case 1–4 to provide insight into the performance of the framework,

Table 3

AIUA results at pedestrian level. Red line indicates the plane.

	Evolutionary period	Number of points*	AIUA	Lowest wind speed [m/s]	Percentage of points with acceptable wind speed
Case study	N/A	6916	2570.65	0.21	18.79
Case 1	17	6717	2163.23	0.34	23.67
Case 2	20	6638	2104.35	0.33	28.43
Case 3	23	6655	2101.40	0.33	27.6
Case 4	25	6560	2087.69	0.34	27.67

*As the urban area densities differ for Cases1–4, the number of points is not the same for all cases.

**Fig. 13.** Comparing building heights variation for (a) Case 2, (b) Case 3, and (c) Case 4.

including the lowest wind speed at the pedestrian level and the percentage of points with the acceptable wind speed. As shown in Table 3, Cases 1–4 (which are the optimal solutions for the 17th, 20th, 23rd, and 25th iterations, respectively) report better figures compared to the original case study. In Table 3, the best values for each factor are highlighted (using gray color background).

According to Table 3, the highest number for the lowest wind speed belongs to Case 1 and 4. While Case 1 is not the optimal case presented in this study (based on the AIUA at the pedestrian level), it performs as well as Case 4 in maximizing the wind speed. Furthermore, considering the figures for the percentage of points with the acceptable wind speed, it is observed that Case 2 reports a better figure than Case 4. The reason for this superiority is the impact of the GA and the PSO operators. By comparing Case 2 and Case 3 (Fig. 13a and 13b), it can be seen that Case 3 has moved beyond the local optimum under the impact of the mutation operator. Indeed, the superiority of Case 3 is due to the impact of the mutation operator in the GA part.

Comparing Case 2 and Case 4 (Fig. 13a and 13c), the relative changing is not dramatic and it should be when the mutation occurs. Thus, no mutation is implemented in Case 4. In addition, more than one

building within the area is changed and offspring is normally differed only in one dimension compared to its parents, and hence, Case 4 is not also the offspring of Case 2. Indeed, Case 2 obtains the optimum under the impact of the PSO part, which results in Case 4.

In general, Case 1–4 are obtained from a specific part of the HGP. The variation in Cases 1–4 confirms the necessity of developing this framework with the help of HGP, not one of the GA or PSO separately.

For further investigation, the AIUA is also calculated for Case 1–4 in a vertical plane within the urban area and between two rows of the buildings. For Case 1, the AIUA in the vertical plane is 189. The figure for Case 2, 3, and 4 is 260, 254, and 199, respectively. Although the best AIUA for the pedestrian level is related to Case 4 (Table 3), the order of the cases is disturbed in higher levels (Table 4). It means that the best AIUA is that of Case 4 in the horizontal plane at the pedestrian level, while Case 1 holds the best AIUA in the vertical plane.

The AIUA for all cases, whether at the pedestrian level or in the vertical plane, is better than the figures for the case study. As shown in Table 4, all four cases report a better status than the case study in all three factors. It means that, wind conditions improvement at the pedestrian level can significantly improve the wind conditions in the

Table 4

AIUA results in vertical plane. Red line indicates the plane.

	Evolutionary period	Number of points	AIUA	Lowest wind speed [m/s]	Percentage of points with acceptable wind speed
Case study	N/A	2376	375.00	0.21	61.03
Case 1	17	2376	189.66	0.26	76.48
Case 2	20	2376	260.63	0.28	70.06
Case 3	23	2376	254.24	0.28	69.85
Case 4	25	2376	199.74	0.29	67.42

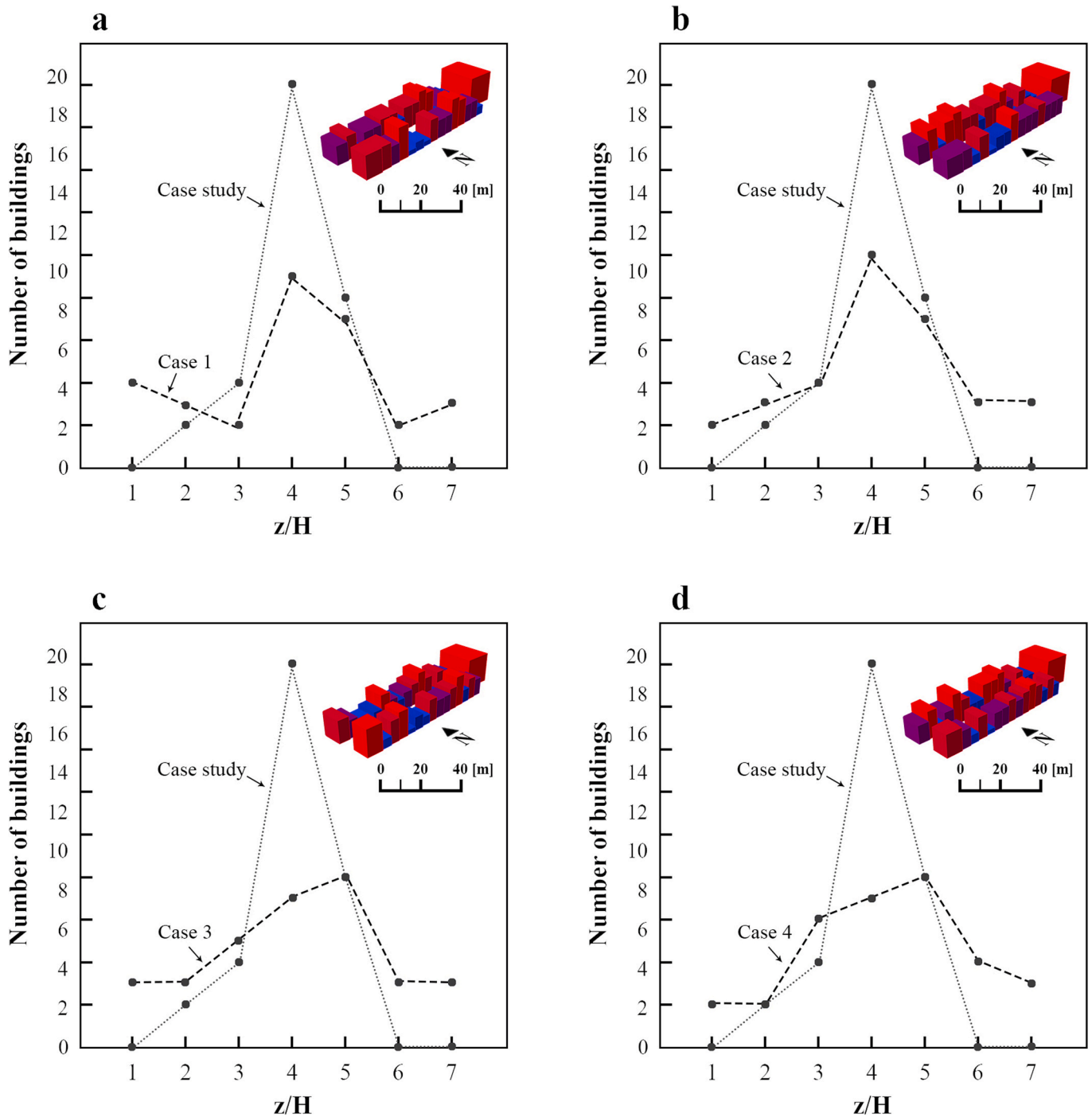


Fig. 14. Comparison of building heights for case study with (a) Case 1, (b) Case 2, (c) Case 3, and (d) Case 4.

whole area.

According to Table 4, the highest wind speed in the vertical plane belongs to Case 4, which is the optimal urban area. However, in the case of the percentage of points with optimal wind speed as well as the AIUA, Case 1 reports the best figures. Therefore, from the standpoint of the AIUA in the vertical plane, Case 1 is the optimal solution. As mentioned earlier, there has been no accurate solution to such complicated problems, which are influenced by several variables. Furthermore, the role of EAs is to approach an optimal solution, not to an exact one.

Although there is no restriction on controlling the amount of total area of living, this area experiences a slight variation of $\pm 200 \text{ m}^2$ in Cases 1–4 when compared to the original case study. Table 5 shows the total area of living in each case.

Fig. 14 compares the number of buildings in each height category for the case study and Cases 1–4. Accordingly, height diversity increases over the optimization process. In the original case study, most buildings are 4-story, while buildings are mostly 3, 4, and 5-story in Case 4.

Eq. (7) presents a metric to calculate building height diversity for each case, where h_i is the building height, h_4 is the height of a 4-story

Table 5

Comparison of total area.

	Case study	Case 1	Case 2	Case 3	Case 4
Total area [m ²]	12190.39	12045.74	12221.02	12002.51	12302.74

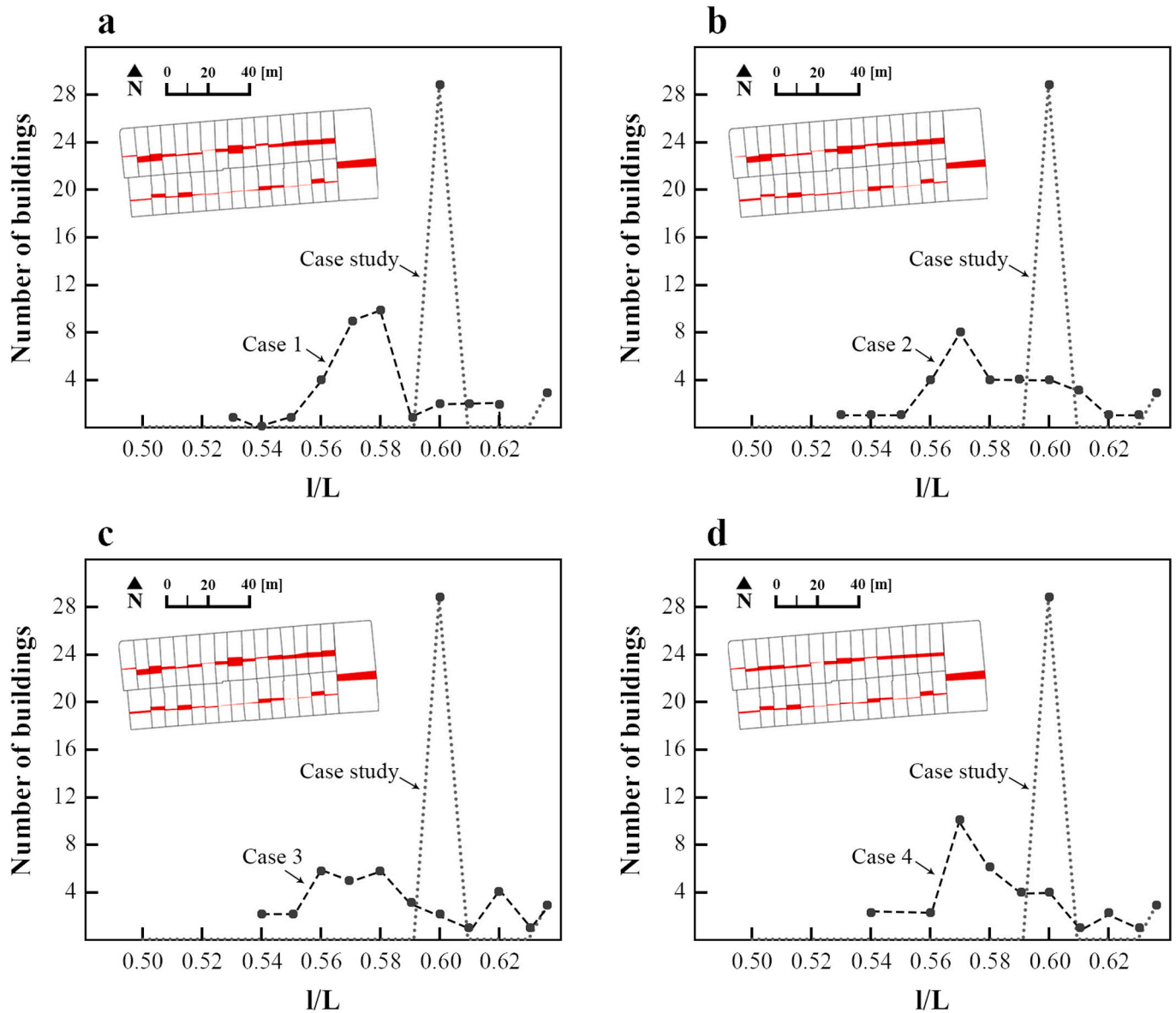


Fig. 15. Comparison of plan area density for case study with (a) Case 1, (b) Case 2, (c) Case 3, and (d) Case 4, with an indication of area differences in red. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

building (the accepted height based on the regulations), and n is the number of buildings in each case. Accordingly, the building heights diversity for Cases 1–4 are 1.25, 1.18, 1.36, and 1.25, respectively. This value is equal to 0.5 for the case study and indicates that the current status of the case study, which holds adjacent buildings of almost the same height, is not optimal from the standpoint of improving wind conditions.

$$\sum_{i=1}^n \frac{|h_i - h_4|}{n \times h_4} \quad (7)$$

The plan area densities of the case study and Cases 1–4 are also compared (Fig. 15). In the case study, almost all buildings have the same plan area densities of 60%, while the plan area densities of the buildings in Cases 1–4 are highly varied. The mean plan area density for Cases 1–4 are 57.8%, 58.1%, 58%, and 58.2%, respectively. This value is 60.4% for the original case study. The area differences between the case study and proposed cases are also shown in Fig. 15.

5. Discussion

Some essential points of this study can be further discussed:

- Given the software and hardware constraints, the size of the initial population and the number of evolutionary periods for the case study is considered to be 20 particles, and 25 iterations, respectively. It should be noted that it is very likely to yield better results if the size of the population increases, and the simulations continue beyond 25 iterations. Future research should further investigate the sensitivity of the accuracy of the results based on the number of iterations.
- In this framework, the hyper-parameters are highly dependent on the case under study and available facilities. Building height diversity in the urban area directly affects decision making for assigning building features, and the number of buildings is also exclusive for each case. On the other hand, some situations would happen that a favorable outcome is not obtained when the termination condition is met. The user may decide then to change the termination condition and increase the number of generations. The higher the size of the initial population and the number of iterations, the more accurate the

results will be. However, the computational time and cost should always be considered.

- Because of the complexity of urban geometry and the high computational cost, only one wind direction is considered in this study. Nevertheless, the introduced framework can be applied for all wind directions and all wind speeds as inlet wind flow.
- The wind comfort criterion used in this study is based on Refs. [50, 51]. However, it can be simply adjusted based on users' needs and preferences.
- Buildings are modeled with simplified geometry. However, earlier studies have shown that façade and roof geometrical details can significantly affect near-surface wind flow [63,66,67]. Future research should consider the impact of these small- and large-scales details on wind conditions in urban areas.
- A threshold of the mean wind speed of 5 m/s is used that is relevant for wind comfort studies. It should be noted that the developed framework is also capable of considering outdoor thermal comfort for which thermal comfort indices such as the Universal Thermal Climate Index (UTCI) [68] can be implemented. Future work can evaluate the performance of the new framework, while both wind and thermal comfort are considered simultaneously.
- The validation study and grid-sensitivity analysis are performed only once and before the optimization starts. However, if changes in geometry are significant during the optimization process, it is recommended to perform grid-sensitivity analysis for more cases to ensure that the CFD results are independent on the grid.

6. Conclusions

This study presents the evaluation of a new framework, which intends to significantly improve wind conditions in complex real urban areas by employing Artificial Intelligence techniques. The combined impact of building height and plan area density variations on wind conditions is studied through four steps, which are (i) urban area identification, (ii) optimization, (iii) CFD simulation, and (iv) evaluation. Such a framework can be a valuable addition to the literature. It can be utilized at the early stages of design by urban planners, architects, and building engineers to improve wind conditions in complex real urban areas. Note that the CFD simulations should be based on a detailed validation and grid-sensitivity analysis.

The proposed framework is implemented in a real urban area, which consists of 32 residential buildings, to improve wind conditions and reduce low-wind-speed regions at the pedestrian level. Thus, the performance of the framework is evaluated by CFD simulations based on the 3D steady RANS equations, and by comparing the results for the case study and optimal cases. In this regard, 920 cases are generated through evolutionary periods using the HGP algorithm. The following calculations are made:

- An Enormous number of simulations of possible combinations to parametric study is required that is, in most cases, very difficult or even impossible. Evolutionary-based algorithms can effectively reduce computational time and significantly improve the wind conditions in the entire urban area.
- Wind conditions improvement at the pedestrian level can lead to improving the wind conditions in the whole area effectively.
- The simulation is performed on a personal computer (x 64 architecture). A 4-core node (Intel(R) Core(TM) i7 6700 HQ CPU @ 2.60 GHz) with 32 GB of system memory is used for the simulation. In total, it takes twelve and a half days to complete such simulation using a personal computer with mentioned properties. Thus, this framework can be easily implemented for complex real urban areas.
- After the simulation is done for the case study, the wind speed is improved by about 19% compared to the current situation based on AIUA. It can be concluded that in each urban area with specific wind speed and direction, a specified number of buildings, and a particular

total area of living, wind conditions can be efficiently improved by properly varying the building heights and the plan area densities. Thus, contrary to what is commonly occurred in cities, it is practically possible to assign specific patterns of building heights and plan area densities to each urban area with an approach to have better wind conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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